

## Week 1

### Program 1-

```
rehan_lenovo@Rehan-Lenovo x + v
GNU nano 8.7.1
#include<stdio.h>
int main(){
printf("Welcome to operating Systems\n");
return 0;
}|
```

Output:

```
rehan_lenovo@Rehan-Lenovo x + v
rehan_lenovo@Rehan-Lenovo:~/week1$ gcc hello.c -o hello
rehan_lenovo@Rehan-Lenovo:~/week1$ ./hello
Welcome to operating Systems
rehan_lenovo@Rehan-Lenovo:~/week1$ |
```

### Program 2-

```
rehan_lenovo@Rehan-Lenovo x + v
GNU nano 8.7.1
#include<stdio.h>
int main(){
char name[30];
printf("Enter your name: ");
scanf("%s",name);
printf("Welcome %s\n",name);
return 0;
}
```

Output:

```
rehan_lenovo@Rehan-Lenovo x + v
rehan_lenovo@Rehan-Lenovo:~/week1$ gcc name.c -o name
rehan_lenovo@Rehan-Lenovo:~/week1$ ./name
Enter your name: Rehan
Welcome Rehan
rehan_lenovo@Rehan-Lenovo:~/week1$ |
```

### Program 3-

```
rehan_lenovo@Rehan-Lenovo X + v
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>

int main(){
    printf("Process ID=%d\n", getpid());
    return 0;
}
```

### Output:

```
rehan_lenovo@Rehan-Lenovo X + v
rehan_lenovo@Rehan-Lenovo:~/week1$ nano processId.c
rehan_lenovo@Rehan-Lenovo:~/week1$ gcc processId.c -o processId
rehan_lenovo@Rehan-Lenovo:~/week1$ ./processId
Process ID=1653
rehan_lenovo@Rehan-Lenovo:~/week1$ |
```

### Program 4-

```
rehan_lenovo@Rehan-Lenovo X + v
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>

int main(){
    fork();
    fork();

    printf("Hello\n");

    return 0;
}
```

### Output:

```
rehan_lenovo@Rehan-Lenovo X + v
rehan_lenovo@Rehan-Lenovo:~/week1$ nano fork.c
rehan_lenovo@Rehan-Lenovo:~/week1$ gcc fork.c -o fork
rehan_lenovo@Rehan-Lenovo:~/week1$ ./fork
Hello
Hello
Hello
Hello
rehan_lenovo@Rehan-Lenovo:~/week1$ |
```

### Program 5-

```
rehan_lenovo@Rehan-Lenovo x + v
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;
    pid=fork();

    if(pid==0)
    {
        printf("I am Child Process\n");
    }
    else{
        printf("I am Parent Process\n");
    }

    return 0;
}
```

Output:

```
rehan_lenovo@Rehan-Lenovo x + v
rehan_lenovo@Rehan-Lenovo:~/week1$ nano parent.c
rehan_lenovo@Rehan-Lenovo:~/week1$ gcc parent.c -o parent
rehan_lenovo@Rehan-Lenovo:~/week1$ ./parent
I am Parent Process
I am Child Process
rehan_lenovo@Rehan-Lenovo:~/week1$ |
```

### Program 6-

```
rehan_lenovo@Rehan-Lenovo x + v
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>

int main(){
    pid_t pid;
    pid=fork();

    if(pid==0){
        printf("Child PID=%d\n", getpid());
        printf("Parent PID=%d\n", getppid());
    }
    else{
        printf("Parent PID=%d\n", getpid());
    }

    return 0;
}
```

Output:

```
rehan_lenovo@Rehan-Lenovo: ~/week1$ nano parent1.c
rehan_lenovo@Rehan-Lenovo: ~/week1$ gcc parent1.c -o parent1
rehan_lenovo@Rehan-Lenovo: ~/week1$ ./parent1
Parent PID=1809
Child PID=1810
Parent PID=1809
rehan_lenovo@Rehan-Lenovo: ~/week1$ |
```

Program 7-

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>

int main(){
    pid_t pid;
    pid=fork();

    if(pid==0)
    {
        printf("Child Running\n");
    }
    else{
        wait(NULL);
        printf("Parent Running After Child\n");
    }

    return 0;
}
```

Output:

```
rehan_lenovo@Rehan-Lenovo: ~/week1$ nano wait.c
rehan_lenovo@Rehan-Lenovo: ~/week1$ gcc wait.c -o wait
rehan_lenovo@Rehan-Lenovo: ~/week1$ ./wait
Child Running
Parent Running After Child
rehan_lenovo@Rehan-Lenovo: ~/week1$ |
```